# JobGenAI – A Generative AI Powered Job Recommendation System

**Project Team**

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**Under the Guidance of:**

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## 1. Introduction

The rapid expansion of online recruitment platforms such as LinkedIn and Naukri has significantly increased access to employment opportunities. However, this growth has also introduced challenges for job seekers, including information overload and inefficient job matching. Most existing job portals rely heavily on keyword-based search mechanisms, which fail to understand the contextual meaning of a candidate's resume, such as actual skills, experience level, and career aspirations.

As a result, job seekers often receive irrelevant job recommendations and are required to spend considerable time manually filtering through job listings. There is a growing need for intelligent systems that can understand resumes contextually and provide meaningful job recommendations.

JobGenAI aims to address this challenge by leveraging Generative Artificial Intelligence and Large Language Models (LLMs) to analyze resumes and recommend relevant job opportunities based on semantic understanding rather than simple keyword matching.

## 2. Problem Statement

Despite the availability of numerous job portals, job seekers continue to face difficulty in identifying opportunities that truly match their skills and

experience. Existing systems lack contextual understanding of resumes and rely on rigid keyword-based matching, which results in irrelevant job suggestions. Manual job searching is time-consuming and does not provide insights into skill gaps or future career preparation.

There is a need for an AI-driven job recommendation system that can intelligently analyze resumes, identify skill gaps, and recommend suitable jobs based on semantic and contextual understanding.

## 3. Objectives

The primary objectives of the JobGenAI project are:

- To analyze resumes using Generative AI techniques
- To extract skills, experience, and career-related information from resumes
- To identify skill gaps and missing competencies
- To recommend relevant jobs from multiple platforms
- To reduce manual effort in job searching

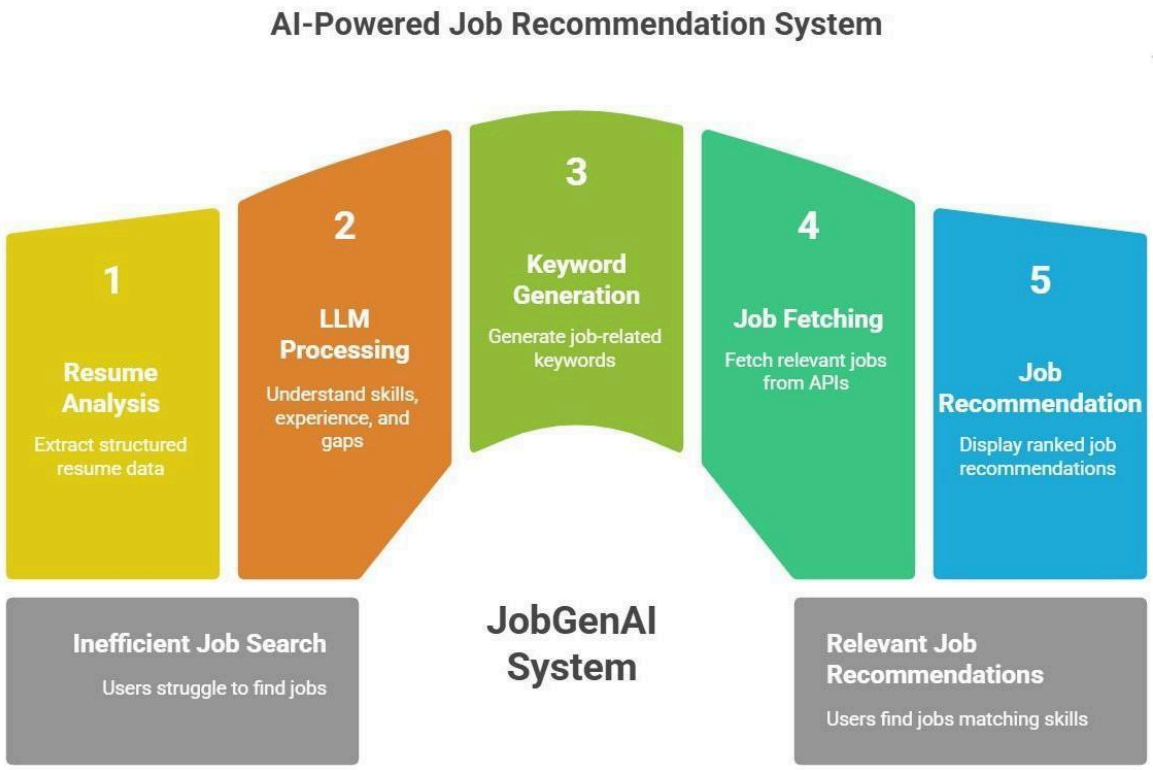
## 4. Proposed System

JobGenAI is a Generative AI powered job recommendation system that accepts a user's resume in PDF format and extracts both structured and unstructured data. A Large Language Model is used to analyze the resume content and generate a summary, identify skills and experience, detect skill gaps, and suggest a future learning roadmap and interview preparation strategy.

Based on the AI analysis, job-related keywords are generated and used to fetch real-time job listings from platforms such as LinkedIn and Naukri using APIs. The system filters and ranks job recommendations and presents them to the user along with application links.

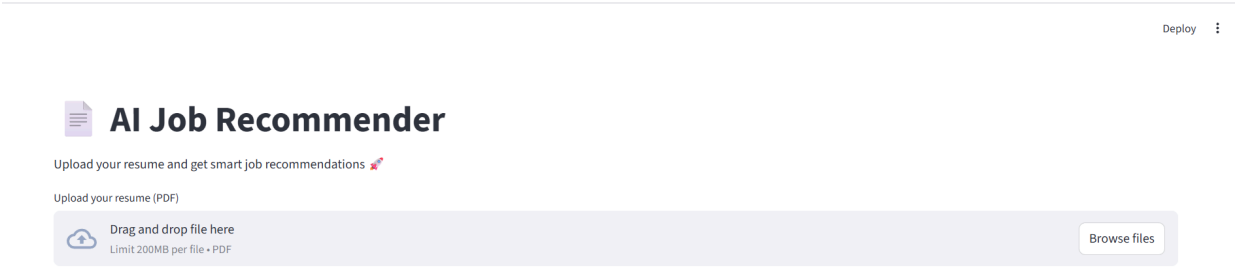
## 5. System Architecture

The system architecture represents the overall workflow of the JobGenAI system, starting from resume upload and ending with job recommendations. It includes modules for resume processing, AI-based analysis, job keyword generation, job fetching from external platforms, and recommendation ranking.

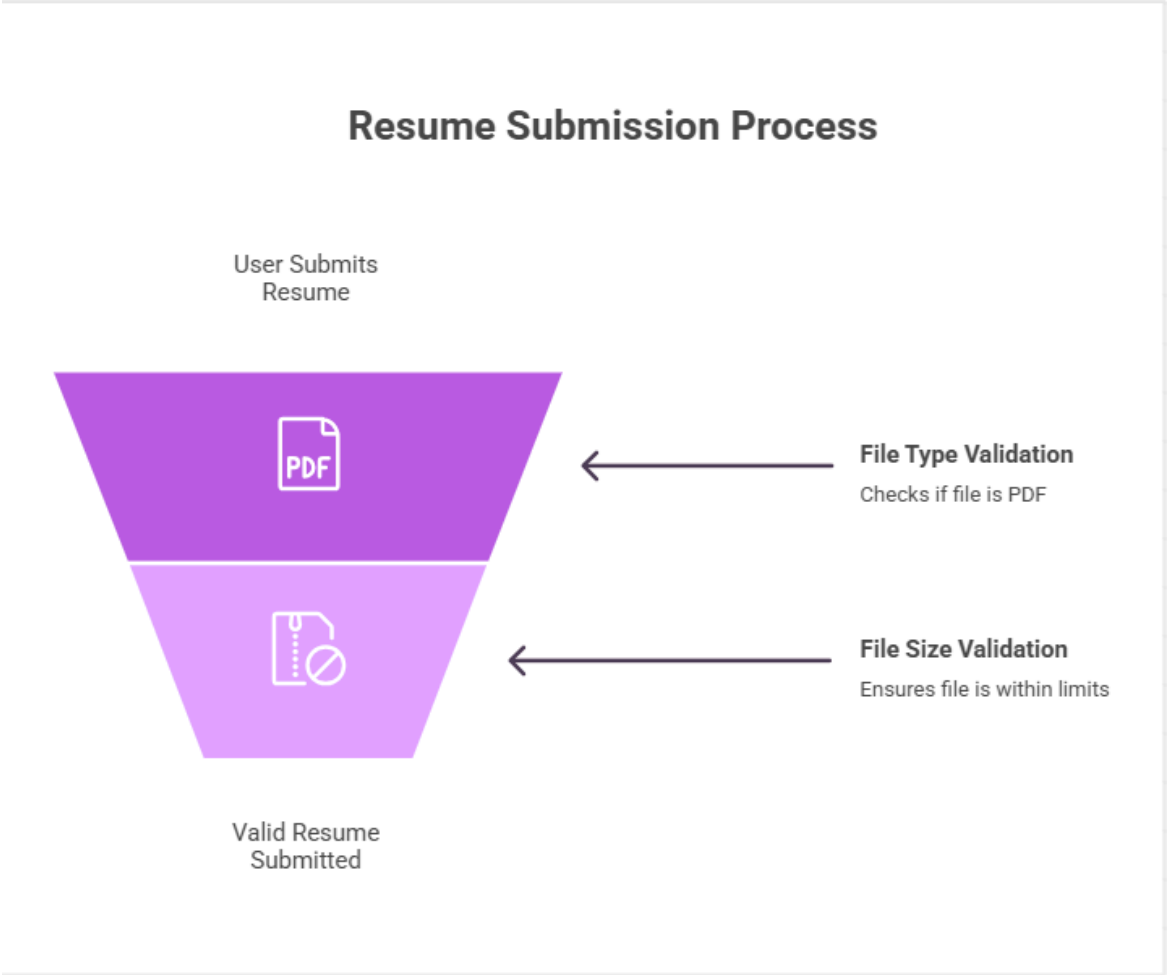



## 6. Module Description

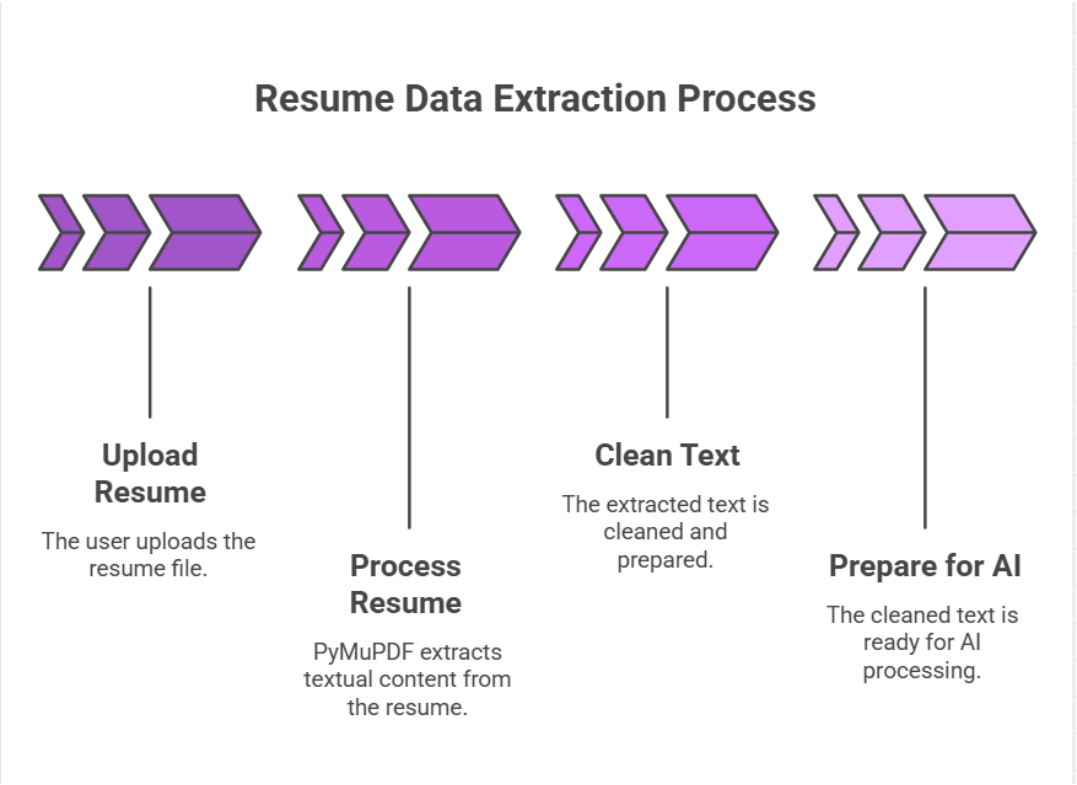
### 6.1 Resume Input Module




This module allows users to upload resumes in PDF format. Basic validation is performed to ensure correct file type and size.

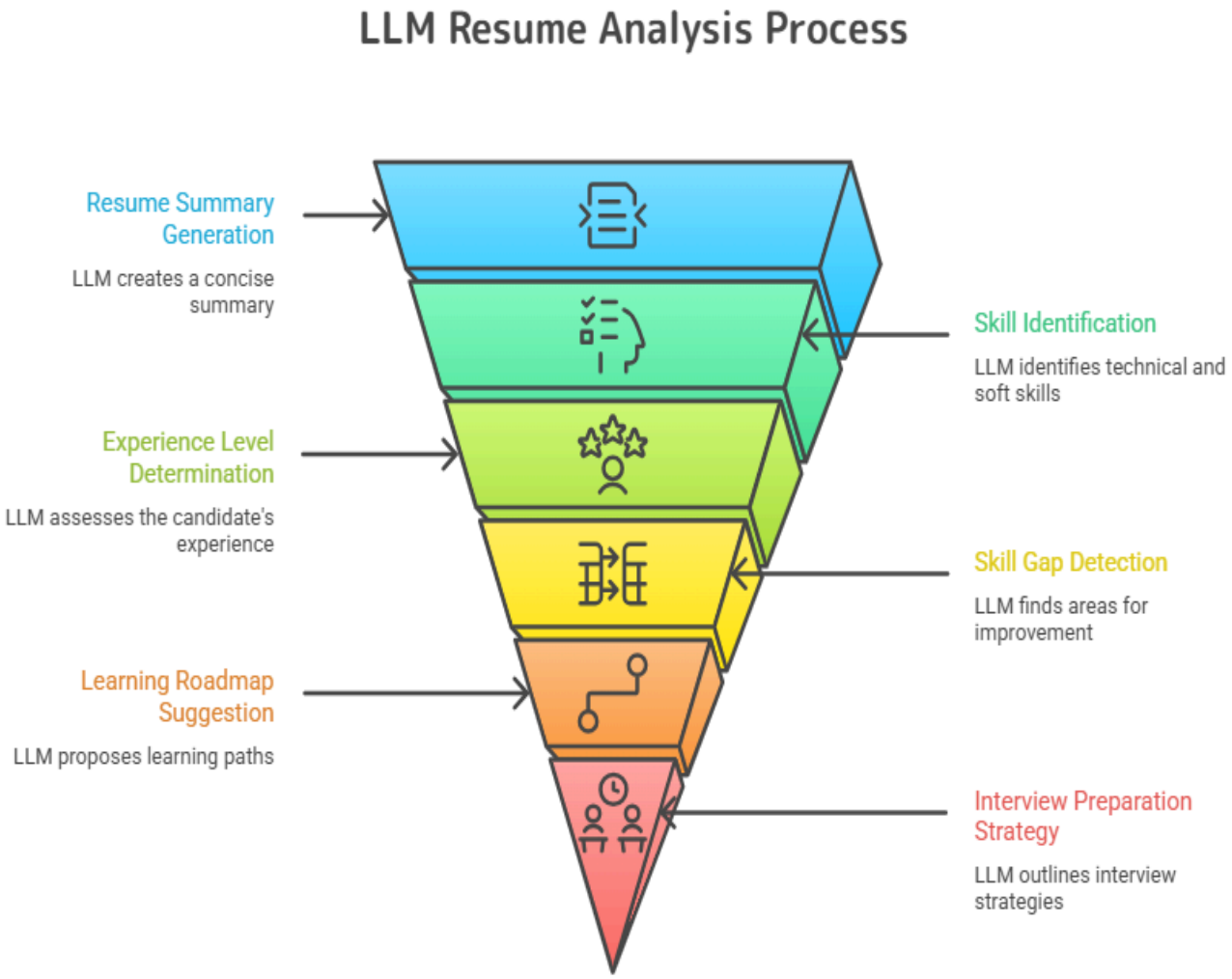



## 6.2 Resume Data Extraction Module



The uploaded resume is processed using PyMuPDF to extract textual content. The extracted text is cleaned and prepared for AI processing.

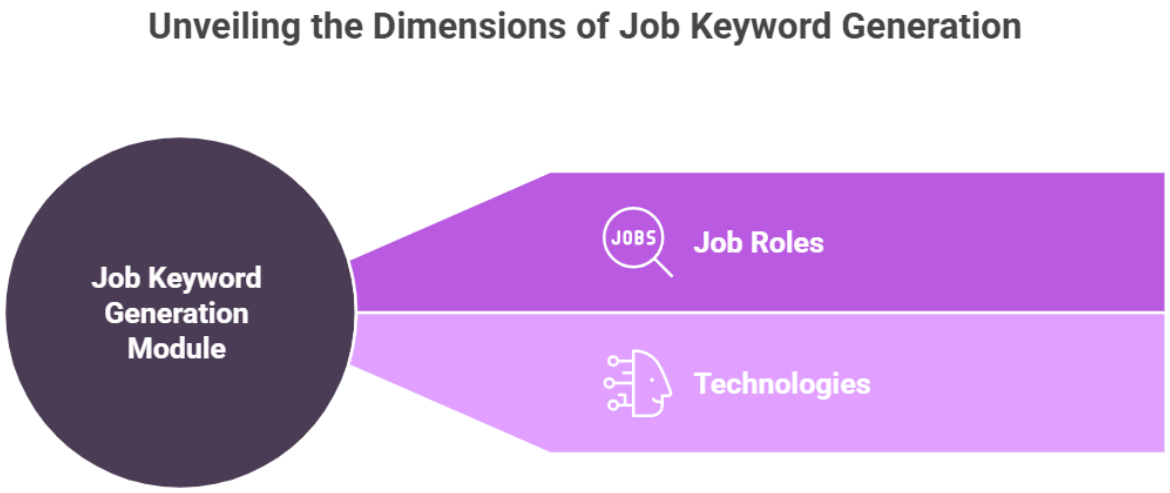
### 6.3 LLM Processing Module




A Large Language Model analyzes the extracted resume text to generate a resume summary, identify technical and soft skills, determine experience level, detect skill gaps, and suggest a future learning roadmap and interview preparation strategy.

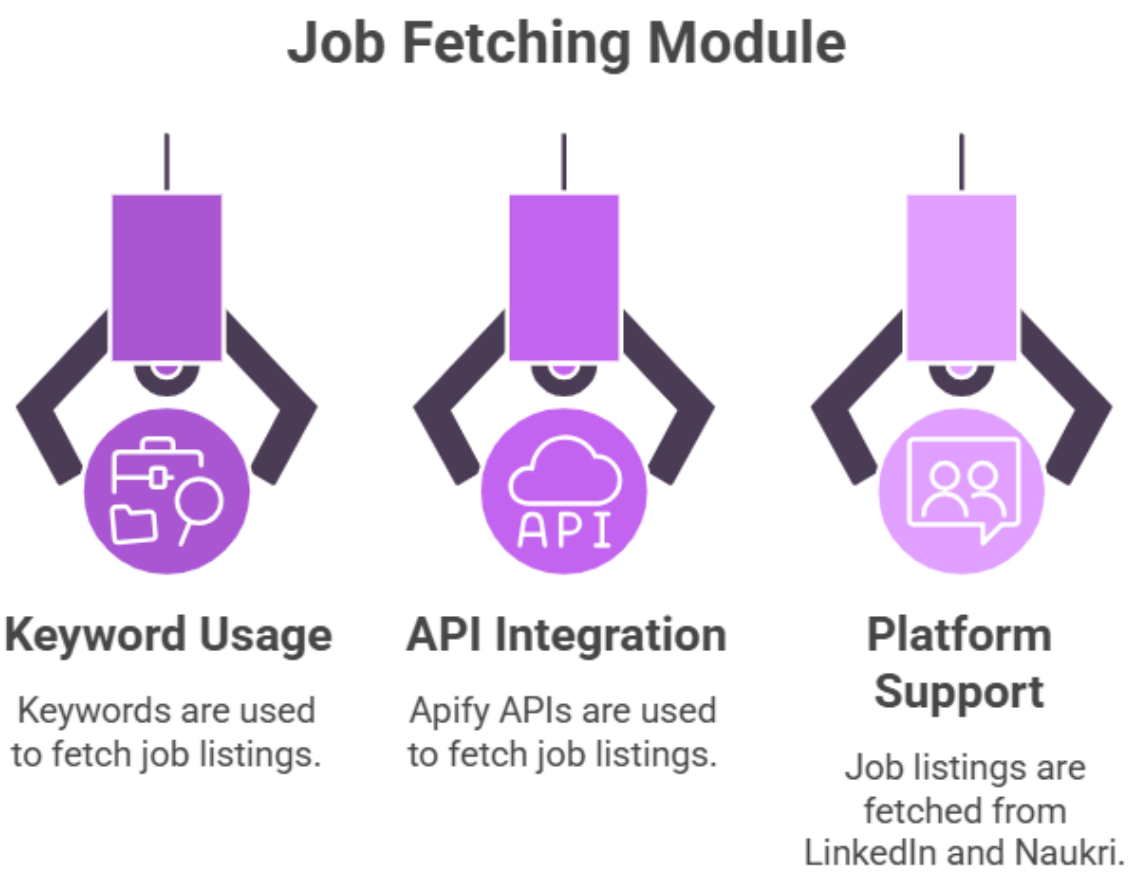
### 6.4 Job Keyword Generation Module

Based on the AI analysis, job-specific keywords are generated, such as job roles and required technologies.



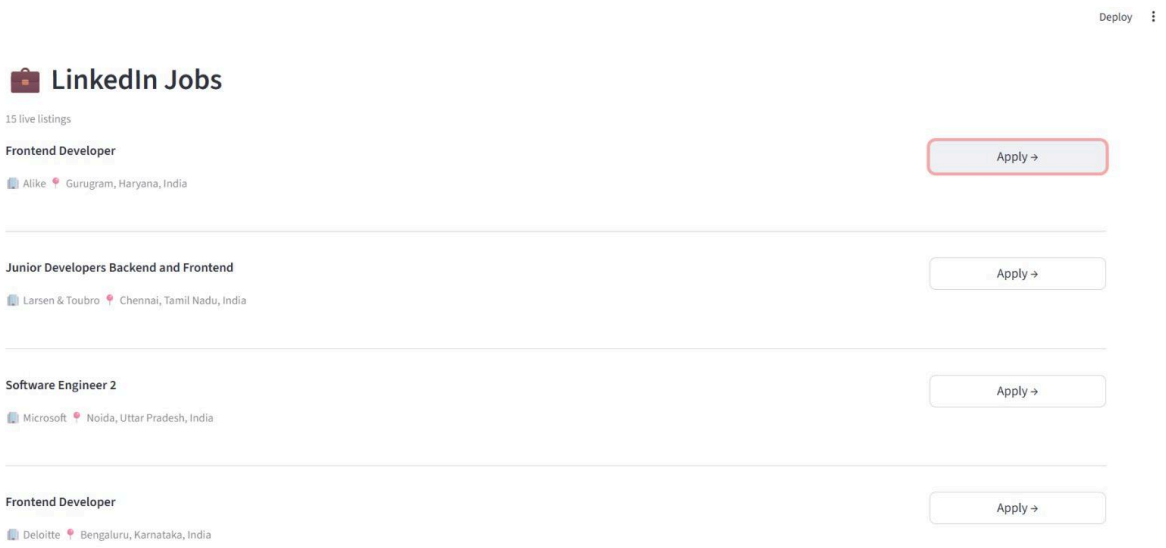

### 6.5 Job Fetching Module

The generated keywords are used to fetch job listings from LinkedIn and Naukri through Apify APIs.




### 6.6 Job Recommendation Module

Fetched job listings are filtered and ranked based on relevance before being displayed to the user.




## 7. Technology Stack

- Programming Language: Python
- Frontend: Streamlit
- Backend: Python-based API architecture
- AI and NLP: GrokAI / Large Language Models
- Resume Processing: PyMuPDF
- Job Data Sources: Apify API (LinkedIn, Naukri)
- Environment Management: python-dotenv

**8. Type of Architecture**

The system follows a modular and scalable architecture, where each component operates independently. This design improves maintainability and allows future enhancements without affecting the entire system.

## 9. Scope of the Project

- Resume analysis and understanding
- Skill gap identification
- AI-generated career roadmap
- Multi-platform job recommendations

## 10. Limitations

- Dependency on third-party APIs
- API rate limits
- Job availability controlled by external platforms

## 11. Future Enhancements

- User authentication and profile management
- Job alert notifications
- Resume ATS score analysis
- Fine-tuned language models for improved accuracy
- Job tracking and application dashboard
- React-based frontend

## 12. Implementation Plan (3 Months)

**Month 1**

Requirement analysis, Python fundamentals, resume upload functionality, resume text extraction, and basic user interface development.

**Month 2**

Integration of Large Language Model, resume analysis, skill extraction, skill gap identification, and job keyword generation.

**Month 3**

Integration of job APIs, job ranking and filtering, user interface improvements, testing, documentation, and final presentation.

## 13. Conclusion

JobGenAI demonstrates the practical application of Generative AI in solving real-world challenges faced by job seekers. By combining AI-driven resume understanding, API integration, and modular system design, the project provides meaningful job recommendations and career insights. This project is suitable for academic evaluation and offers strong industry relevance.